

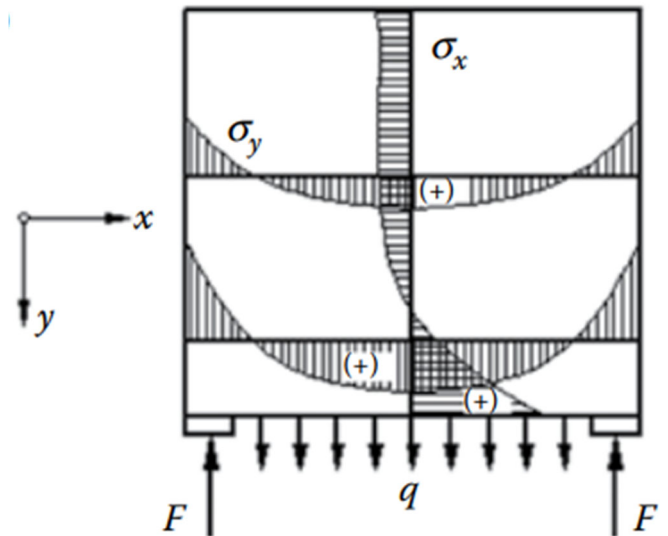
Note: Assignment 5 is due on Friday 29 May 2026, 3:00 pm. No late submission without prior approval from the lecturer is accepted.

Assignment 5 – Q.1: (30 marks)

Develop a suitable strut-and-tie model for each of the following cases. Give detailed justification for your developed strut-and-tie model.

- (i) A simply supported beam carrying a uniformly distributed load (Figure 1). (10 marks)

Figure 1. A simply supported beam carrying a uniformly distributed load.



- (ii) A simply supported deep beam with opening (Figure 2). The beam width is 400 mm. (20 marks)

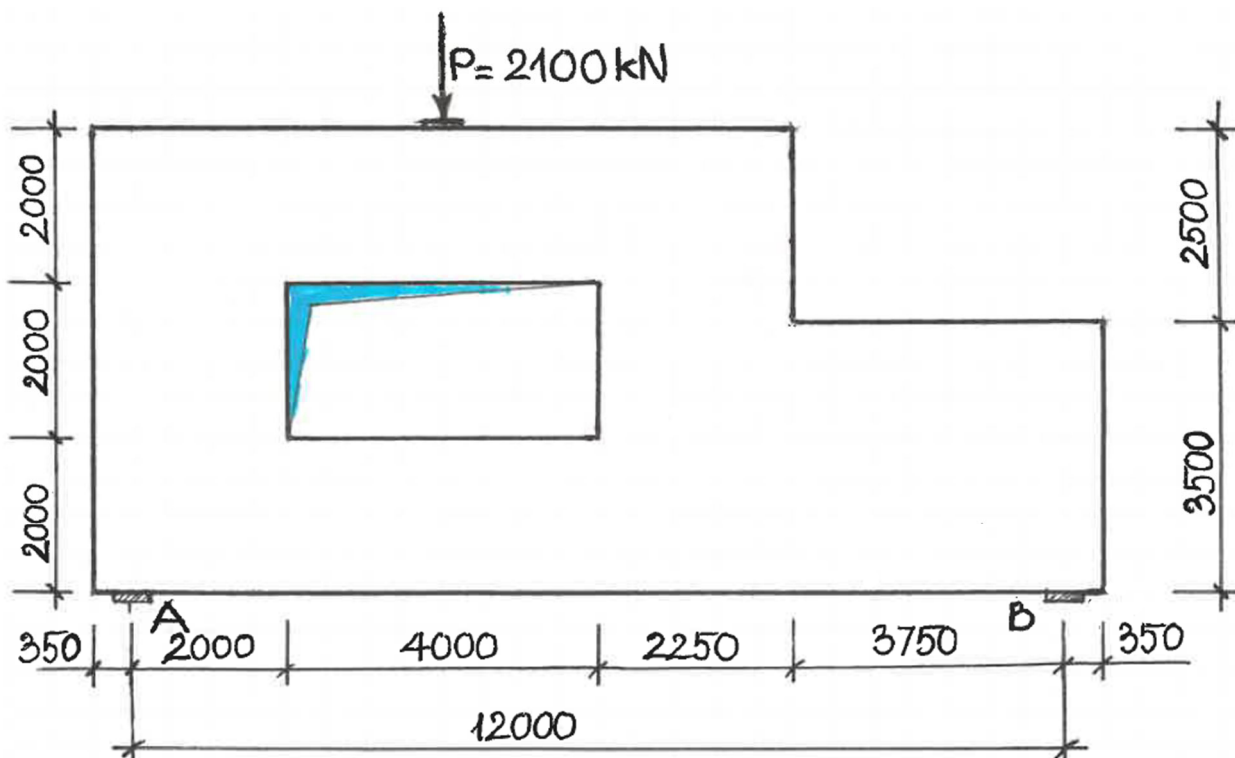


Figure 2. A simply supported deep beam with opening and varying depth.

Assignment 5 – Q.2: (70 marks)

On the basis of an appropriate strut-and-tie model, design the deep beam (Figure 3). Relevant details as follows:

- The width of all members (including beam and columns) is 600 mm.
- The factored design load $P = 4 \times m + n$ (kN); where m and n are the first 3 digits and the last 2 digits of your 8-digit student number, respectively. For example, if your student number is 49012345, your P value will be: $P = 4 \times 490 + 45 = 2005$ kN.
- $f'_c = 50$ MPa; $f_{sy} = f_{sy.f} = 500$ MPa.
- Clear cover to reinforcement = 30 mm.
- Using relevant clauses in AS3600-2018.
- If your student number is an even number, your allocated support in questions **c-d** is the left support. If your student number is an odd number, your allocated support in questions **c-d** is the right support.

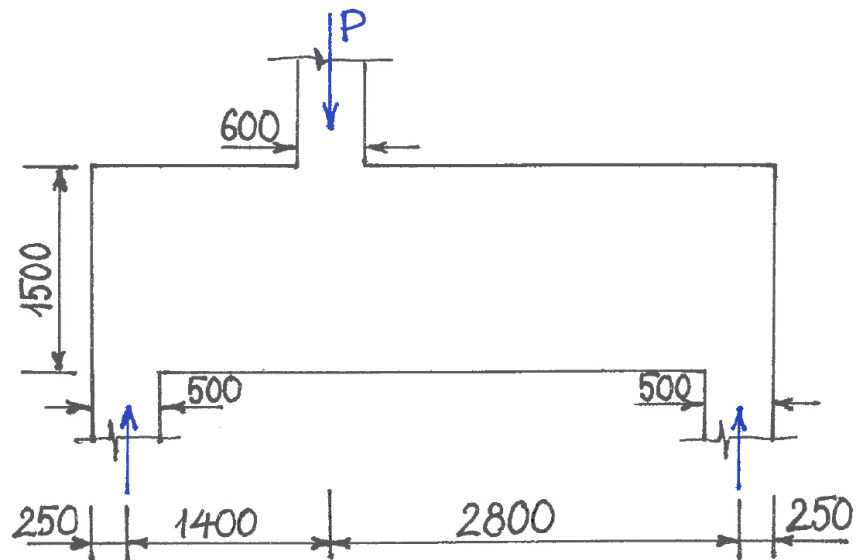


Figure 3. A simply supported beam carrying a point load.

You are required to:

- a. Identify the B- and D- regions. Develop a suitable strut-and-tie model. Give justification for your chosen strut-and-tie model. (15 marks)
- b. Select steel reinforcement for each of the tension ties. (10 marks)
- c. Check stress levels in all struts and in the node at your allocated support. Using Mohr's circles to determine the maximum compressive stress in your allocated nodal zone when checking the stress level in that node. (30 marks)
- d. Design for bursting reinforcement for the inclined strut coming into your allocated support. Assuming the strut force at serviceability limit state is 0.6 of that at ultimate limit state. (15 marks)

Solution Guide:**Assignment 5 – Q.1:** (30 marks)

Develop a suitable strut-and-tie model for each of the following cases. Give detailed justification for your developed strut-and-tie model.

- (i) A simply supported beam carrying a uniformly distributed load (Figure 1). (10 marks)

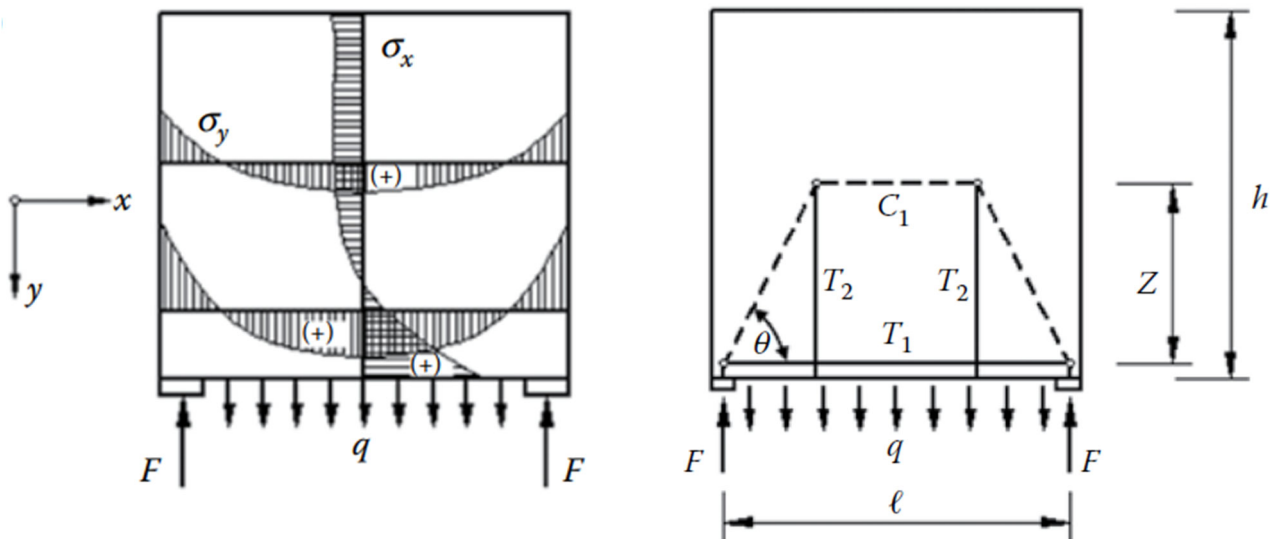


Figure 1. A simply supported beam carrying a uniformly distributed load.

- When formulating a suitable STM for this, the following considerations should be taken into account:
 - The STM should be developed using a combined approach, integrating: the elastic stress distribution, the load-path method, and the expected natural structural behaviour of the beam. These approaches help ensure that the STM realistically represents the flow of internal forces through the structure.
 - Based on the combined consideration of the elastic stress distribution and the natural behaviour of the beam, the horizontal tension tie T_1 should be located at approximately the centroid of the tensile stress block in the mid-span section. This positioning allows the tie to represent the corresponding principal tensile force path within the member.
 - Similarly, using the combined insights from the elastic stress field and the load-path method, the locations of the vertical tension ties T_2 can be established.
 - The compressive member C_1 should be positioned at approximately the centroid of the compressive stress block in the mid-span section, thereby representing the corresponding principal compression force path through the beam.
 - The proposed STM should be checked to ensure that: all external loads and reactions are in equilibrium; a sufficient number of struts and ties are provided to maintain equilibrium at each node.
 - As a final verification, the model should be reviewed to confirm that the assumed load paths are consistent with the expected stress trajectories and that the resulting reinforcement arrangement is practical and constructible. For good serviceability and crack control, reinforcement should generally follow the natural flow of tensile forces within the structure.

- (ii) A simply supported deep beam with opening (Figure 2). The beam width is 400 mm. (20 marks)

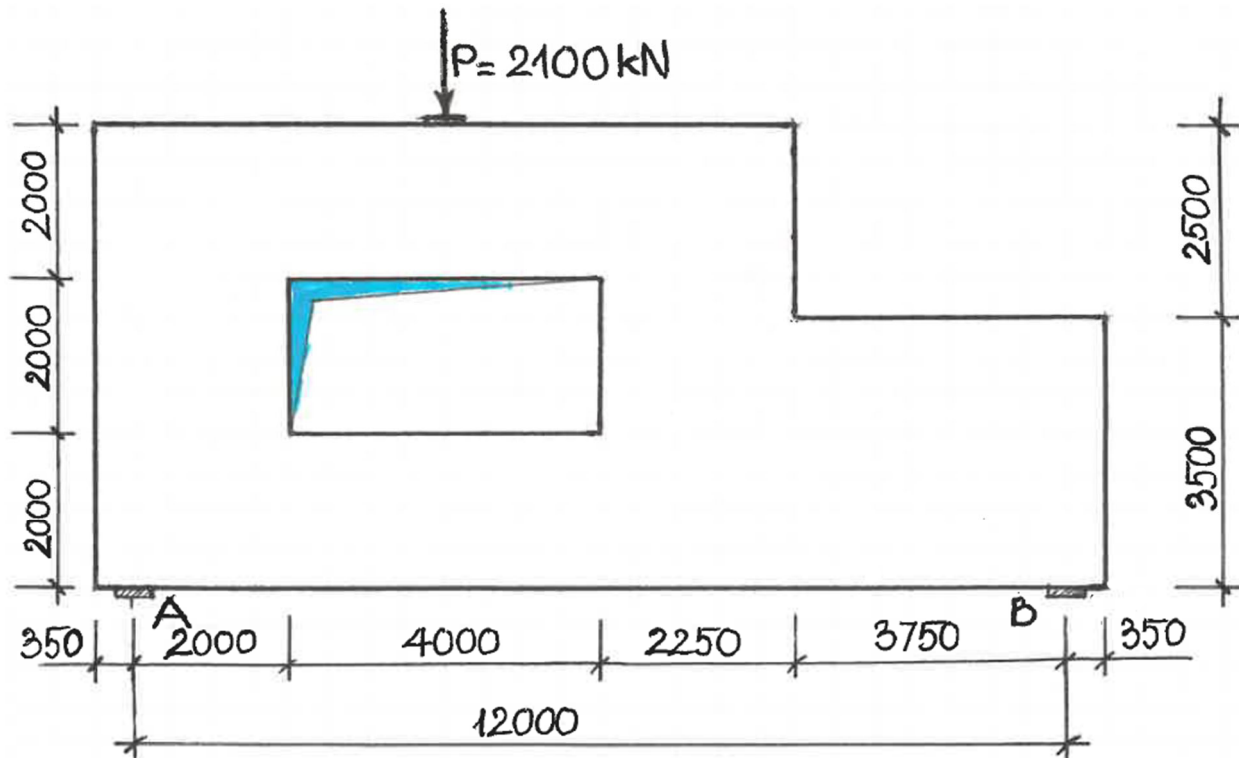
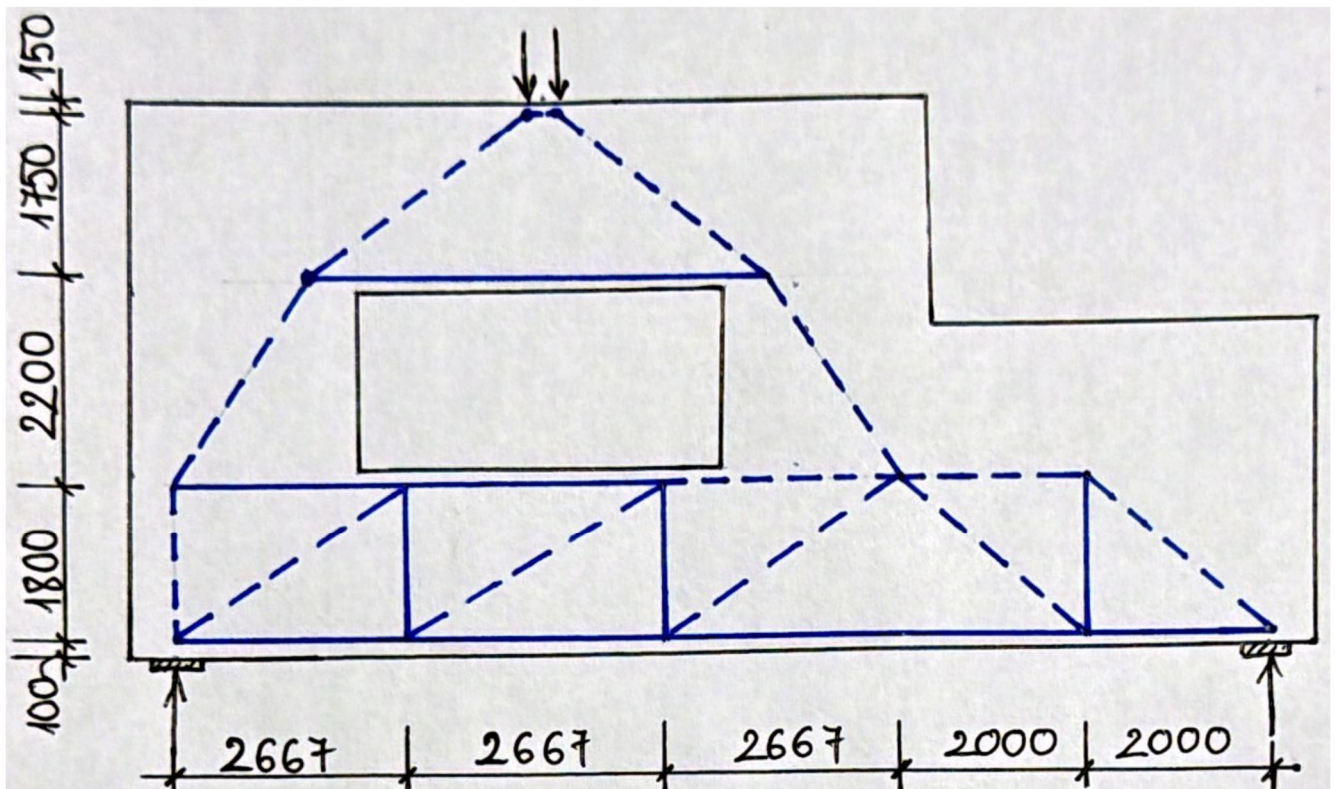


Figure 2. A simply supported deep beam with opening and varying depth.

- The structure can be modelled as an upper deep beam spanning on two inclined columns, which are in turns supported on a lower deep beam. Based on the expected behaviour, a suitable STM can be developed using the combined approach.



Assignment 5 – Q.2: (70 marks)

a. **Identify the B- and D- regions. Develop a suitable strut-and-tie model. Give justification for your chosen strut-and-tie model.** (15 marks)

- **Identification of B- and D-regions:** Applying St Venant's principle, it is evident that the influence zones associated with the concentrated load and support reactions extend over the whole member. Consequently, the strain distribution throughout the beam is expected to be significantly nonlinear, and the entire structure may reasonably be treated as a D-region (disturbed region). Accordingly, the strut-and-tie method (STM) is adopted for the design and assessment of the whole structure.
- **Development and Justification of the Strut-and-Tie Model**

The proposed STM is developed based on a combined consideration of the load-path method, the shear-force diagram, the expected natural structural behaviour, and the practical arrangement of reinforcement.

Key considerations in developing the model are as follows:

- The applied load of $P = 2100$ kN is divided into support reactions of 1400 kN and 700 kN at the left and right supports, respectively. These support reactions establish the primary force-transfer mechanism through the structure.
- Direction of inclined struts 1-5, 2-5 and 3-4 should be consistent with shear-force diagram.
- The geometry of the STM is established by assuming: Node 5 at depth of $600/4 = 150$ mm from the top face; Tension ties 1-2 and 2-3 at 150 mm from the bottom face; remaining node locations as shown in the figure, while ensuring that all strut-to-tie angles satisfy the recommended minimum angle of 30° .
- The proposed STM should be checked to ensure that: all external loads and reactions are in equilibrium; a sufficient number of struts and ties are provided to maintain equilibrium at each node.

The resulting STM therefore provides a rational representation of the load-transfer mechanism, satisfies equilibrium requirements, and is consistent with both the anticipated stress field and practical reinforcement detailing considerations.

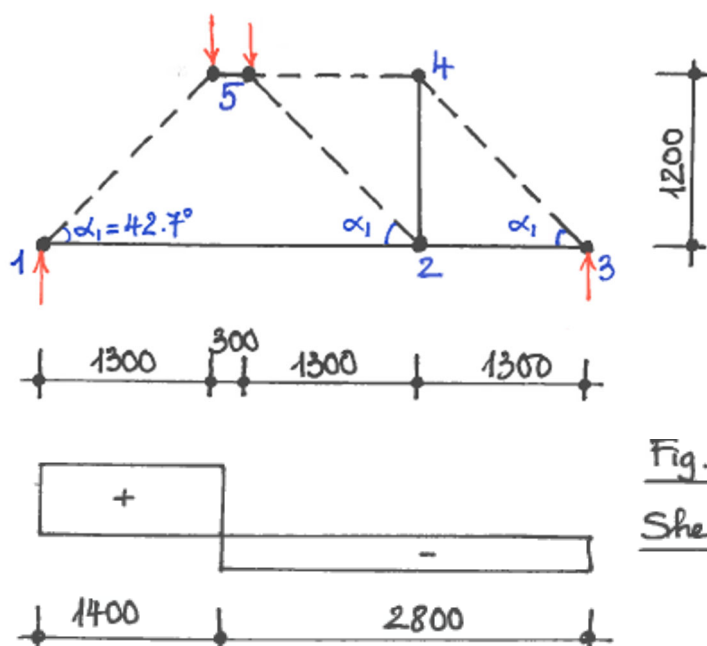


Fig. 1.
Shear force diagram

b. Select steel reinforcement for each of the tension ties.

Tie 1-2:

- $T_{1-2} = 1517.2 \text{ kN}$
- Required A_{st} : $A_{st} = 1517.2 \times 10^3 / (0.85 \times 500) = 3569.8 \text{ mm}^2$.
Choose 12N20 ($A_{st} = 3768 \text{ mm}^2$) to be distributed in 4 layers of 3N20 each, with the centroid of the reinforcement located approximately 150 mm above the bottom face, consistent with the assumed location of Tie 1–2 in the strut-and-tie model.

Tie 2-3:

- $T_{2-3} = 758.6 \text{ kN}$
- Required A_{st} : $A_{st} = 758.6 \times 10^3 / (0.85 \times 500) = 1785 \text{ mm}^2$.
Choose 6N20 ($A_{st} = 1884 \text{ mm}^2$) to be distributed in 2 layers of 3N20 each, with the resultant tensile force at 150 mm from the bottom face - consistent with the location of tension tie initially assumed.

Tie 2-4:

- $T_{2-4} = 700 \text{ kN}$
- Required A_{st} : $A_{st} = 700 \times 10^3 / (0.85 \times 500) = 1647.1 \text{ mm}^2$.
Choose 8 sets of 3-legged N10 at 170 mm spacing ($A_{st} = 1872 \text{ mm}^2$) to be symmetrically distributed around the assumed location of Tie 2-4.

c. Check stress levels in Strut 1-5 and in Node 1 at the left support.

- ❖ Force in strut 1-5: 2064.4 kN.
- ❖ Limiting stress in Strut 1-5: $\phi_c \times \beta_s \times 0.9 \times f'_c = 0.65 \times 0.5634 \times 0.9 \times 50 = 16.48 \text{ MPa}$
 - $\beta_s = 1/[1+0.66\cot^2(42.7^\circ)] = 0.5634$
- ❖ Limiting stress in the node: $\phi_c \times \beta_n \times 0.9 \times f'_c = 0.65 \times 0.8 \times 0.9 \times 50 = 23.4 \text{ MPa}$
 - $\beta_n = 0.8$ for CCT node.
- ❖ The strut stress:

$$\frac{2064.4 \times 10^3}{(300 \times \cos 42.7^\circ + 500 \times \sin 42.7^\circ) \times 400} = 9.02 \text{ MPa}$$

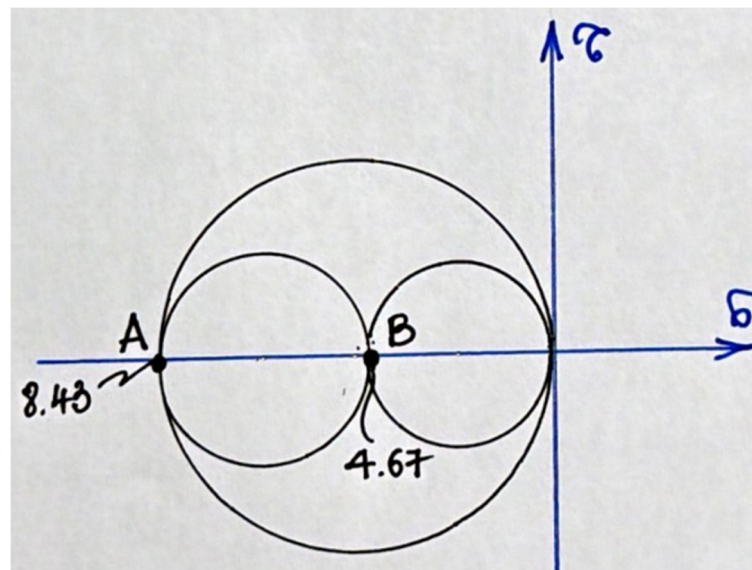
which is significantly less than 16.48 MPa → Great!

- ❖ To check that the limiting stress in the node is not exceeded:
 - For this node, as the stresses on the bottom and left nodal faces are perpendicular to the corresponding nodal face, the maximum compressive nodal stress is the higher of these two stresses – And this maximum compressive nodal stress must be limited to the limiting stress in the node. That means:

$$\frac{1517.2 \times 10^3}{300 \times 600} = 8.43 \text{ MPa}$$

$$\frac{1400 \times 10^3}{500 \times 600} = 4.67 \text{ MPa}$$

- ❖ The maximum compressive nodal stress is thus 8.43 MPa, significantly less than 23.4 MPa
→ Great!
- ❖ The Mohr's circle representing stress state in Node 1 must pass through **Point A** (representing the compressive stress acting on the left nodal face) and **Point B** (representing the compressive stress acting on the bottom nodal face). Since the stresses acting on both faces are purely normal stresses (with no shear stress component), Points A and B lie on the horizontal σ -axis of the Mohr's circle. Consequently, the line segment AB forms a diameter of the Mohr's circle, which completely defines the stress state within the nodal zone. The maximum compressive stress is thus 8.43 MPa.



- d. Design for bursting reinforcement for the inclined strut coming into your allocated support. Assuming the strut force at serviceability limit state is 0.6 of that at ultimate limit state.*

For the strut under consideration:

$$C = 2064.4 \text{ kN (for ULS)}$$

$$C_s = 0.6 \times 2064.4 = 1238.6 \text{ kN (for SLS)}$$

Corresponding bursting forces:

$$T_b^* = 2064.4/5 = 412.9 \text{ kN}$$

$$T_{b.s}^* = 1238.6/2 = 619.3 \text{ kN}$$

$$T_{cr}^* = 0.7 b l_b f_{ct} = 0.7 \times 600 \times 1186.1 \times 2.546 = 1268.1 \text{ kN}$$

$$0.5 T_{cr}^* = 634.1 \text{ kN (> } T_b^* \text{ and > } T_{b.s}^*)$$

- $d_c = \sqrt{(300^2 + 500^2)} = 583.1 \text{ mm}$
- $l_b = \sqrt{(1300^2 + 1200^2)} - d_c = 1186.1 \text{ mm}$
- $f_{ct} = 0.36\sqrt{50} = 2.546 \text{ MPa}$

- ❖ Bursting reinforcement is not required for this strut.